

Ganzorig Gankhuyag

Efficient & Edge AI | Autonomous-Driving Perception | Vision-Language Models On-Device

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PROFESSIONAL SUMMARY

Ph.D. researcher with 8+ years at KETI working on both **computer-vision perception** and **efficient on-device AI** - building state-of-the-art vision and vision-language models and making them run inside real latency, memory and power budgets on edge hardware: mobile and automotive NPUs (Qualcomm Hexagon/Snapdragon, NextChip, Synaptics), embedded ARM and automotive SoCs (TI TDA4VM, NVIDIA Jetson/Xavier), and Raspberry Pi.

Perception. Real-time semantic segmentation (77.1% mIoU on Cityscapes at 102 FPS), 2D and 3D object detection (YOLOv7, BEVFusion, PointPillar), traffic-light recognition, and a full in-cabin sensing system: driver gaze (0.956 GazeFollow AUC at 15M parameters, real-time on-device), occupant action recognition (94.12% top-1, 9 classes), and vision-language seat-state monitoring.

Efficiency. INT8 post-training quantization, quantization-robust reparameterization (RepConv), structural pruning, and TensorRT / ONNX / LiteRT deployment, with a 2x parameter reduction at a 0.02 mAP cost, and 2x faster on-device LLM inference at 4.91 W vs. 5.52 W. Three first places and two second places across international efficiency and model-optimization challenges (LPCVC 2026, MAI 2025, AIS 2024, Mobile AI 2022, AI Grand Challenge 2020). 637 citations, h-index 10, four U.S. patents (three granted).

EDUCATION

Ph.D., Electrical & Electronic Engineering - Yonsei University 2011 - 2020

Image & Information Lab. Coursework and dissertation completed 2016; degree conferred 2020.
Dissertation: A Study on Finite Field Compressive Sensing Framework for Network Coding.

M.S., Electronics & Information Communication - Konkuk University 2006 - 2009

VLSI Lab.
Dissertation: Implementation of an H.264 encoder using Intel Wireless MMX technology.

B.S., Engineering in Information & Communication - Huree University of ICT, Mongolia 2002 - 2006

HONORS & AWARDS

1st Place, Track 3 - Low-Power CV Challenge, mobile-VLM AIGC (Qualcomm Snapdragon)	LPCVC 2026 @ CVPR	2026
Outstanding Employee Award	Korea Electronics Technology Inst. (KETI)	2026
2nd Place - Quantized Image Super-Resolution	MAI 2025 @ CVPR	2025
1st Place - Real-Time Image Super-Resolution	AIS 2024 @ CVPR	2024
1st Place - Real-Time Super-Resolution (Team Leader)	Mobile AI & AIM 2022 @ ECCV	2022
2nd Place - AI Model Optimization (Object Detection)	AI Grand Challenge, IITP	2020
Best Paper Award	IICCC, IPACT	2015

SKILLS & LANGUAGES

Efficient & Edge AI: TensorRT, ONNX Runtime, CUDA, INT8 PTQ/QAT, Structural Pruning, TFLite/LiteRT; NVIDIA Jetson/Xavier, TI TDA4VM, NextChip NPU, Qualcomm Hexagon/Snapdragon NPU, Hailo, Embedded ARM, Raspberry Pi

Computer Vision: Semantic Segmentation (UNet, HardNet), 2D Detection (YOLOv7/v8/v11, ViT), 3D Detection (BEVFusion, PointPillar, Apollo platform), Action Recognition (pose/GNN, R(2+1)D, LSTM), Gaze Estimation, Super-Resolution, Image Restoration

Deep Learning: PyTorch, TensorFlow, CLIP, DINOv3, DETR, Transformers, CNNs, GNN, Vision-Language Models, LoRA fine-tuning (Qwen2-VL, LLaMA)

Video / Codecs: AV1 (tiled encoding), HEVC (H.265), H.264/AVC, MPEG-DASH

Programming & Tools: Python, C/C++ (embedded), MATLAB; Linux, ROS / ROS 2, Git, Docker, FFmpeg, GStreamer, OpenCV

Languages: Mongolian (Native), Korean (Fluent - TOPIK Level 6), English (Fluent)

EMPLOYMENT HISTORY

Korea Electronics Technology Institute (KETI)

Jun 2018 - Present

Postdoctoral Researcher | South Korea

R&D Project: Efficient Single-Image Super-Resolution

- **CASR: Efficient 4K Real-Time Super-Resolution.** Proposed the CASR model with a new advanced RepConv block and a channel-aware network structure. Won 1st place at AIS 2024 (CVPR) with 33.11 dB PSNR (QP31) at a 0.47 ms GPU runtime.
- **Lightweight Super-Resolution for Edge Devices.** Proposed an efficient SR network with a Reparameterization (RepConv) method that reduces INT8 quantization error. 1st-place solution at the Mobile AI & AIM 2022 Real-Time Single-Image Super-Resolution Challenge (ECCV) with 30.03 dB PSNR at a 19.20 ms runtime on the Synaptics VS680 edge NPU (Team Leader).

R&D Project: On-Device LLM Inference

- **Efficient LLM Inference on Edge Devices.** Designed a lightweight INT8 (LiteRT) inference pipeline for deploying LLaMA-3.2, Qwen, and DeepSeek on the Raspberry Pi 4, achieving 2x faster inference latency and reduced power (4.91 W vs. 5.52 W) versus framework-native PyTorch baselines while maintaining output quality (ICCV Workshop 2025).

R&D Project: Generated-AI Image / Content Evaluation

- **Mobile-VLM AIGC Detection (On-Device Authenticity).** Built a two-stage vision-language detector on Qwen2-VL-2B (LoRA) within a hard 500-token-per-stage budget on a Snapdragon 8 Gen 5: Stage 1 produces free-text reasoning over eight visual criteria, Stage 2 emits a structured JSON decision; an auxiliary classification head added a +2.08 pp gain on the 26,033-image Chameleon benchmark. Won 1st place in the 2026 Low-Power Computer Vision Challenge (LPCVC) Track 3 at CVPR 2026, with an official rubric score of 0.827 at 30.48 TPS.
- **Multimodal Quality Assessment of AI-Generated Content (ITFMN).** Proposed the ITFMN network, a multimodal architecture that learns features from both the text prompt and the generated image, scoring 86% on the NTIRE 2024 Quality Assessment of AI-Generated Content dataset.

R&D Project: Driver State Monitoring

- **Driver-State Monitoring System for UN-R171 DCAS.** Turned the UN-R171 Section 5.5.4 driver-state monitoring requirement into a testable specification, defined and collected the in-cabin dataset, proposed the recognition methods, and implemented the running system from camera driver and ROS pipeline through to the output signals. Gaze zone, eye closure (EAR), PERCLOS and head pose feed a six-rule Euro NCAP state machine over NORMAL / DISTRACTED / DROWSY / UNRESPONSIVE, every threshold kept in a config file so the rules stay easy to check against the standard. Mapped how the system meets UN-R171, GSR II, Euro NCAP and UN-R157.
- **Gaze Estimation Component (QueryGaze).** Single-pass gaze-target model reusing DETR detection queries as person-specific gaze prompts on a frozen DINOv3 backbone, deployed to an automotive edge NPU with ViT-aware INT8 quantization. 0.956 GazeFollow AUC (above the 0.924 human benchmark) at 15M parameters (3-60x smaller than competing models), in real time on-device; validated with multi-seed, placebo-controlled ablations and zero-shot cross-domain evaluation across four gaze benchmarks.

R&D Project: In-Cabin Sensing System - Driver & Occupant Monitoring

- **Occupant Action & State Recognition.** Two-branch action model behind the deployed system (CNN video branch plus keypoint branch, concatenated then linearly classified) at 94.12% validation top-1 across 9 action and state classes. Separately: INT8 quantization rescales each branch's logits independently and breaks logit-space fusion, so designed an architecture-level late-fusion rule for a two-branch R(2+1)D-18 ensemble, combining after softmax normalization instead of in logit space, giving higher accuracy on a Qualcomm Hexagon NPU at no measurable latency cost (IEEE Access, under review).
- **Multi-Camera In-Cabin Inference System (production path).** Designed the deployed inference system for a national fleet-management R&D program (TRL 4 to 7): four in-cabin cameras publish to four ROS 2 topics on a single NVIDIA Jetson AGX Orin 64GB, which sends one 10 Hz JSON stream to the vehicle's occupant-monitoring controller over TCP, covering 10 classes per seat for up to 5 occupants. TensorRT FP16 cut in-cabin inference from 14.44 to 9.59 ms per frame while holding the 10 Hz contract, matching the PyTorch results on all 120 seat-state cases. Mixed precision set by measurement: INT8 for detection, FP16 for pose (INT8 cost 38x keypoint error).

- **VLM Left-Object / Seat-State Detection.** Real labeled in-cabin images are scarce, so built a two-stage vision-language pipeline (routing then subtype classification) trained on synthetic labels distilled from a Qwen2.5-VL-72B teacher with match-with-retry filtering, scaling training data 200x (5 real photos to 1,000+ synthetic) via LoRA fine-tuning of Qwen2-VL-2B and a scripted prompt library over the Kling AI kling-v3-omni image-to-image API.

R&D Project: Perception for Autonomous Car

- **2D High-Efficiency Real-Time Semantic Segmentation.** Developed a real-time semantic-segmentation model using a UNet architecture with HardNet blocks and a self-attention module, and built an end-to-end framework with TensorRT inferencing ported to ROS. Achieved 77.1% mean IoU on the Cityscapes dataset at 102 FPS on an RTX 2080Ti GPU.
- **Traffic-Light Classification & Recognition.** Researched and implemented traffic-light classification models using ResNet and ResNet+LSTM (90% accuracy), and integrated detection with a YOLOv7 + SORT tracker, classifying over both single and temporal frames.
- **3D Object Detection (Autonomous Test-Driving License).** To prepare for autonomous test-driving licensing, implemented a lightweight 3D object detection model using CNN segmentation methods, tailored for the Apollo framework on Baidu's Autonomous Driving Platform.

R&D Project: Safe L4 Autonomous Driving System

- **Camera Soil Detection.** Proposed a camera-based soil detection algorithm for surround-view lenses and optimized a lightweight model for the NVIDIA Xavier embedded environment, keeping real-time speed on edge devices.
- **Error-Pixel Recovery for Automotive Sensors.** Proposed an error-pixel detection and recovery algorithm for embedded devices and optimized a lightweight model for Texas Instruments' AI-embedded boards (TDA4VM) to restore defective pixels at real-time inference speeds.

R&D Project: Road Traffic Network Control in Emergencies and Disasters

- **2D Real-Time Detection & Classification.** Optimized a YOLOv7 detector for emergency-vehicle and traffic-police perception, applied structural network pruning for a 2x parameter reduction with only a 0.02 mAP drop, and deployed to the NextChip APACHE6 NPU (95% accuracy at 45 FPS).
- **Police Hand-Gesture Action Recognition.** Proposed a Linear Transformer network fusing visual appearance with keypoint data, reaching 99% accuracy at 33 FPS. Designed the deployed inference system: camera ingest through a ROS 2 package, on-site inference on an edge compute box at a traffic control tower, and recognition results published to other systems over an API. Reached real-time throughput through efficient frame buffering and reduced CPU workload.

R&D Project: 4K Tiled Video Streaming of 16K 360-degree Contents

- **Motion-Constrained AV1 Encoder for Tiled Distributed Transcoding.** Researched and implemented motion-constrained AV1 encoding so tiles remain independently decodable for distributed transcoding, including MV search-range and loop-filter (deblocking, CDEF, loop restoration) restriction algorithms, plus multiple encoder speed-up techniques.

Chowis Co., Ltd.

Nov 2016 - May 2018

Computer Vision Researcher | South Korea

- **Skin & Hair Diagnosis Image Analysis.** Developed skin-pore, wrinkle, and hair-loss detection algorithms for handheld diagnostic devices (over 90% diagnostic accuracy) and set up standard camera-lens and LED tuning, and image acquisition QA processes.

Yonsei University - Image & Information Lab

Sep 2011 - Oct 2016

Graduate Research Assistant (Ph.D.) | South Korea

- **Next-Generation DTV Broadcasting (Content Production / Transmission / Quality Evaluation).** Developed an adaptive bitrate-selection scheme for guaranteed QoS in MPEG-DASH streaming, an enhanced CDN + peer-to-peer hybrid network architecture, and RTSP-based verification for the MPEG-DASH server.
- **Low-Complexity Video Stitching for Personalized UI/UX.** Built syntax-level, high-speed image stitching using H.264 flexible macroblock ordering (FMO) and an alternative hybrid-structure approach without FMO; ported a tiled H.264 encoder in FFmpeg to the VLC player.

Huree University of ICT

Mar 2009 - Aug 2010

Assistant Professor, Electronic Engineering Department | Ulaanbaatar, Mongolia

- Taught undergraduate courses in the Electronic Engineering Department.

- **Efficient Video Compression for Embedded Systems.** Ported Windows Mobile (WinCE) firmware directly to a PCB board and developed an H.264 encoder acceleration algorithm ported to an SoC board.

SELECTED PUBLICATIONS (637 CITATIONS, H-INDEX 10 - FULL LIST ON GOOGLE SCHOLAR)

Under review (first author)

- **G. Gankhuyag**, et al., "Late-Fusion Dual-Path Inference: Architecture-Level Quantization Robustness for INT8 Action Recognition on Mobile NPUs," IEEE Access. (In-cabin occupant action recognition; late fusion after softmax normalization instead of logit-space fusion, on a Qualcomm Hexagon NPU.)

Accepted

- J. Jeong, S. Kim, **G. Gankhuyag***, "Modular Spatial Grounding for Sim2Real Traffic-Safety VQA," ECCVW (AI City Challenge), 2026. (*Corresponding author. Poster; Spatial Grounding Adapter routing spatial questions to pose / bird's-eye geometry, 80.63% to 81.61% real-domain VQA.)
- **G. Gankhuyag**, et al., "Motion Trails as Visual Prompts: A Vision-Language Pipeline for Traffic Anomaly Reasoning," ECCVW (AI City Challenge), 2026. (Poster; shared Qwen3-VL-8B with task-routed LoRA; introduces trajectory visual prompting.)

Published

- K. Yoon, ..., **G. Gankhuyag***, J. Jeong, "LiteRT-Optimized INT8 LLM for Raspberry Pi4 Deployment," ICCVW, 2025.
- K. Yoon, **G. Gankhuyag***, et al., "RCENet: Recursive Concatenation and Enhancement Network for Real-Time Super-Resolution," CVPRW, 2025.
- **G. Gankhuyag**, et al., "Lightweight Deep Learning Model for Defective Pixel Detection and Recovery from the Image Sensors," ECCVW, 2024.
- K. Yoon*, **G. Gankhuyag***, et al., "CASR: Efficient Cascade Network with Channel Aligned Method for 4K Real-Time Single Image Super-Resolution," CVPRW, 2024.
- **G. Gankhuyag**, et al., "Unified Lightweight Traffic Police Action Recognition Network for Edge Device," ICTC (Int'l Conf. on ICT Convergence), 2024.
- **G. Gankhuyag**, H. Yae, Y. Shim, C. Park, K. Min, "A Lightweight Traffic Police Action Recognition Deep Learning Network for Edge Device," ICCE-Asia, 2022.
- B.-I. Mun, **G. Gankhuyag**, et al., "Point-Cloud Perspective Transform via Super-Resolution Model," ICCE-Asia, 2022.
- **G. Gankhuyag**, et al., "Skip-Concatenated Image Super-Resolution Network for Mobile Devices," IEEE Access, Vol. 11, 2022.
- A. Ignatov, R. Timofte, ..., **G. Gankhuyag**, et al., "Efficient and Accurate Quantized Image Super-Resolution on Mobile NPUs (MAI & AIM 2022 Challenge Report)," ECCVW, 2022.
- **G. Gankhuyag**, C. Park, K. Min, "High-Efficient Semantic Segmentation for Autonomous Vehicle," KICS, 2021.
- **G. Gankhuyag**, J. Jeong, Y. Kim, "Advanced Motion-Constrained AV1 Encoder for 8K 360 VR Tiled Streaming," ICTC, 2019.
- **G. Gankhuyag**, Y. Choe, "Sparse Recovery using Sparse Sensing Matrix based Finite Field Optimization in Network Coding," IEICE Trans. Information and Systems, 2017.

SELECTED PATENTS (U.S.)

- K. W. Min, **G. Gankhuyag**, et al., "Deep learning-based motion recognition method and system using multiple feature information," US20250200761A1, 2025.
- K. W. Min, **G. Gankhuyag**, et al., "Behavior Recognition AI Network System and Method for Efficient Recognition of Hand Signals and Gestures," US12406532B2, 2025.
- K. W. Min, **G. Gankhuyag**, et al., "Accelerated Processing Method for Deep Learning-Based Panoptic Segmentation Using an RPN Skip Based on Complexity," US12051230B2, 2024.
- Y. H. Kim, **G. Gankhuyag**, et al., "Motion-constrained AV1 encoding method and apparatus for tiled streaming," US11290714B2, 2020.